

Ramped up RAM

	MRAM	FRAM	Ovonic memory	3D memory	DRAM	SRAM	Flash
Density	High	High	Medium	High	Medium	Low	Medium
Technology	Giant Magnetoresistance	Ceramics	Crystalline state change due to heat	3D fabrication and chip design	Silicon charge store capacitance	Silicon transistor flip-flops	Charge storing transistors
Data transfer speed	Very high	High	Medium	High	Medium	Very high	Low
Cost of implementation	High	High	High	Medium	Low	High	Medium
Time to implement	2-5 years	3-5 years	4-5 years	3-5 years	Already present	Already present	Already present
Applications	Desktops/portables	Desktops/portables	Embedded systems	Portables, Flash replacement	Desktops and portable PCs	System and processor cache	Portable digital devices
Companies involved	IBM, Infineon, Motorola	Ramtron, Samsung, NEC, Infineon	Energy Conversion Devices, Inc	Matrix Semiconductor	Kingston, Crucial, Samsung, Hyundai	ST Microelectronics, Samsung	Intel, AMD, Sandisk

present-day volatile memory. Another problem is that the data can be written only once, much like a CD-R disk—there is no way data can be modified in this form of storage.

Smaller chip sizes translate into significantly lower manufacturing costs and this subsequently spells cheaper components to the end-user. Consider this case:

In most chip fabrication processes, there are usually about five bad spots in a silicon wafer. If this wafer were used to produce, say, 10 chips, this would imply that five chips would be faulty and would have to be discarded. However, if this same wafer were used to produce 100 chips, there would still be just five faulty chips in the batch. 3D fabrication will use a very small footprint due to the expansion in height of the fabrication process, and therefore this form of memory is expected to be significantly cheaper than other solid-state storage technologies today.

The next radical memory implementation is Ovonic memory. In this solid-state memory, heat is used to trigger a reaction in certain types of crystalline materials. These crystalline materials can exist in one of two chemical states that correspond to a digital zero or one. The factor that causes these states to change is the heat applied to these crystals. However, the main hurdle here is controlling the voltage in these devices—a state change occurs only when the crystalline material is heated

at about 600 degrees Centigrade, implying heavy power consumption. The upside is that the access time of prototypes of this technology has been logged at about 50 nanoseconds—about the same speed as today's DRAM, but slower than MRAM technology.

New horizons

Each of these new technologies has marked a new trail in the fabrication of volatile memory. With the inherent advantages seen in terms of performance and lowered fabrication costs and the plethora of applications these technologies can be put to, they seem all set to edge out the existing solid-state memories into oblivion.

In terms of performance, technologies such as MRAM have already demonstrated speeds that are magnitudes faster than existing solid-state memory standards. Once fabrication processes and yields get further improved, this type of memory will be seen in PC-based applications where large densities and performance is of paramount importance. Its low power consumption will also enable it to surface in portable and mobile devices like PDAs,

MP3 players and cell phones. Technologies such as 3D memory boast of inherent characteristics such as lower production costs per megabyte and very small footprints. The immediate applications projected by companies involved in these technologies have been in portable applications such as digital camera and PDA memory.

There are a handful of companies that are actively involved in these new memory technologies. Skymoon Ventures, the company that invested in RAM-BUS, is now funding Matrix Semiconductor, a prime pusher of 3D memory. Ramtron has been one of the proponents of the FRAM technology. This company was founded in the 1980s and released its first FRAM chips in 1992. Initial avatars of this technology produced memory chips where data used to degrade over time and could record information about only a dozen times before losing their storage ability. However, the newer iterations are far more stable and have lower manufacturing costs. On the MRAM front, IBM has partnered with Infineon and has already demonstrated chips operating at sub-10 nanosecond speeds.

In order to keep up with the advancements in other areas of computing such as processing and 3D graphics, the need for significantly faster memory that is both denser and lower in cost, is of increasing importance. Companies involved in the development of these technologies are determined to turn high-volume solid-state memory into commodity items found at the same store as your digital camera. This does not seem so far fetched when considering that our SIM cards are nothing but 32 K Flash RAM implementations! We are at the dawn of the revolution in volatile memory and the future can only get faster. 

MARCO D'SOUZA

